Probability and Statistics

Continuous Random Variables and Probability Distributions

(PART 1)

曹圣彬
cccvolcano@m.scnu.edu.cn

華南師範大學

1 Probability density function

2 Cumulative distribution function

3 Expected value

4 Normal distribution

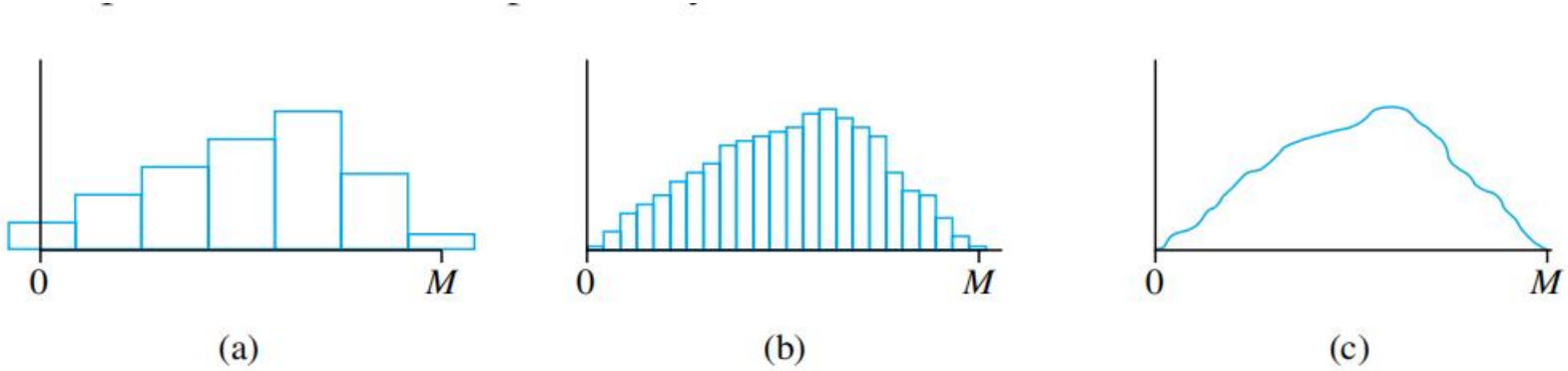
5 Exponential and Gamma distribution

Probability density function

- Continuous random variable （连续随机变量）
- All random variables we have seen so far are discrete ones
  - Possible values constitute a finite set or can be listed one by one
  - The values in counting: number of students, number of gas pumps in use
- Continuous random variable
  - Possible values comprise a single interval or a union of disjoint intervals
  - The values in measurement: height (cm), weight (pound), time (seconds)
  - If a chemical compound is randomly selected and its PH value X is checked, then X is a continuous random variable because

Probability density function

- Probability distribution for continuous random variable
- Suppose the variable X is the depth of a lake. Let M be the maximum depth so that any number in the interval [0, M] is a possible value of X
- The PMF histogram of X measured in meters is shown in Fig. (a)
- The PMF histogram of X measured in centimeters is shown in Fig. (b)
- The PMF histogram of X measured in extreme small unit is shown in Fig. (c)



Probability density function

- Probability distribution for continuous random variable
- The probability distribution function for continuous random variable is called probability density function (PDF), and can be denoted as $f_X(x)$ or $f(x)$
- What is the probability that continuous random variable X is in the range [a, b]?

Probability density function

- Probability distribution for continuous random variable
- The probability distribution function for continuous random variable is called probability density function (PDF), and can be denoted as $f_X(x)$ or $f(x)$
- What is the probability that continuous random variable X is in the range [a, b]?

$$P(a \leq x \leq b) = \int_a^b f_X(x)dx$$

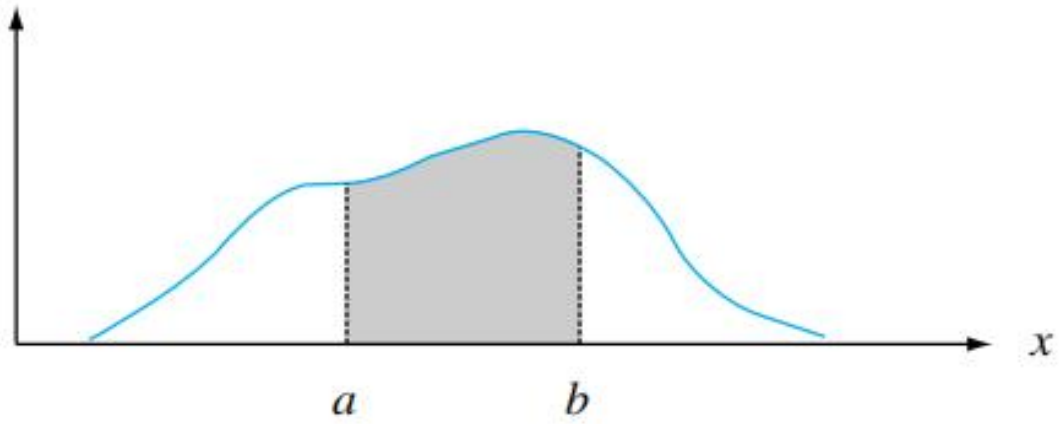
华南师范大學

Probability density function

- Probability distribution for continuous random variable
- The probability distribution function for continuous random variable is called probability density function (PDF), and can be denoted as $f_X(x)$ *or* $f(x)$
- What is the probability that continuous random variable X is in the range [a, b]?

$$P(a \leq x \leq b) = \int_a^b f_X(x)dx$$

- The area under probability density function curve between a and b

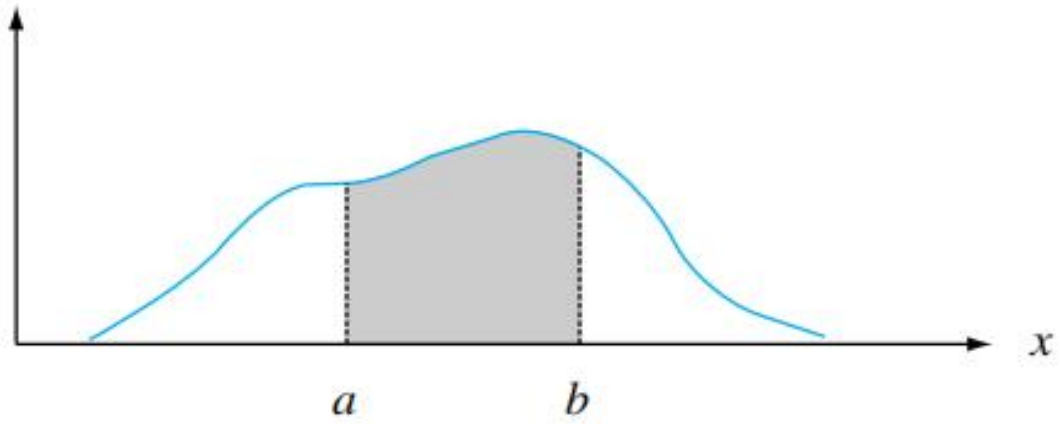


Probability density function

- Probability distribution for continuous random variable
- The probability distribution function for continuous random variable is called probability density function (PDF), and can be denoted as $f_X(x)$ *or* $f(x)$
- What is the probability that continuous random variable X is in the range [a, b]?

$$P(a \leq x \leq b) = \int_a^b f_X(x)dx$$

- The area under probability density function curve between a and b



1. $f(x) \geq 0$ for all x

2. $\int_{-\infty}^{\infty} f(x)\,dx$ = area under the entire graph of $f(x)$ $= 1$

Probability density function

- When X is a discrete random variable, each possible value is assigned positive probability
- This is not true of a continuous random variable because the area under a PDF curve that lies above any single value is zero
- $P(X = c) = 0$, for any possible value c in PDF

Probability density function

- When X is a discrete random variable, each possible value is assigned positive probability
- This is not true of a continuous random variable because the area under a PDF curve that lies above any single value is zero
- $P(X = c) = 0$, for any possible value c in PDF

$$P(X = c) = \int_c^c f(x)\,dx = \lim_{\varepsilon \to 0} \int_{c-\varepsilon}^{c+\varepsilon} f(x)\,dx = 0$$

Probability density function

- When X is a discrete random variable, each possible value is assigned positive probability
- This is not true of a continuous random variable because the area under a PDF curve that lies above any single value is zero
- $P(X = c) = 0$, for any possible value c in PDF

$$P(X = c) = \int_c^c f(x)\,dx = \lim_{\varepsilon \to 0} \int_{c-\varepsilon}^{c+\varepsilon} f(x)\,dx = 0$$

- The probability that X lies in the interval between a and b does not depend on whether a or b is included

$$P(a \le X \le b) = P(a < X < b) = P(a < X \le b) = P(a \le X < b)$$

Probability density function

- Assume that X is a continuous random variable with the following PDF:

$$f_X(x) = \begin{cases} A(2x - x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

- What is the value of A?

Probability density function

- Assume that X is a continuous random variable with the following PDF:

$$f_X(x) = \begin{cases} A(2x - x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

- What is the value of A?

$$\int_{-\infty}^{\infty} f_X(x)dx = \int_{-\infty}^{0} 0dx + \int_{0}^{2} A(2x - x^2)dx + \int_{2}^{\infty} 0dx = \int_{0}^{2} A(2x - x^2)dx = 1$$

Thus, we obtain

$$A\left[x^2 - \frac{x^3}{3}\right]_0^2 = 1$$

$$A\left(4 - \frac{8}{3}\right) = \frac{4A}{3} = 1$$

$$A = \frac{3}{4}$$

Probability density function

- Assume that X is a continuous random variable with the following PDF:

$$f_X(x) = \begin{cases} A(2x - x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

- What is $P(X > 1)$?

Probability density function

- Assume that X is a continuous random variable with the following PDF:

$$f_X(x) = \begin{cases} A(2x - x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

- What is $P(X > 1)$?

$$\begin{aligned} P[X > 1] &= \int_1^{\infty} f_X(x)dx \\ &= \frac{3}{4}\int_1^2 (2x - x^2)dx = \frac{3}{4}\left[x^2 - \frac{x^3}{3}\right]_1^2 \\ &= \frac{3}{4}\left[\frac{4}{3} - \frac{2}{3}\right] \\ &= \frac{1}{2} \end{aligned}$$

Probability density function

- Is the following function a legitimate PDF?

$$f(x) = \begin{cases} \frac{x^2}{9} & 0 \le x \le 3 \\ 0 & \text{otherwise} \end{cases}$$

Probability density function

- Is the following function a legitimate PDF?

$$f(x) = \begin{cases} \frac{x^2}{9} & 0 \le x \le 3 \\ 0 & \text{otherwise} \end{cases}$$

$$\int_{-\infty}^{\infty} f(x)dx = \int_0^3 \frac{x^2}{9}dx = \left[\frac{x^3}{27}\right]_0^3 = 1$$

1. Probability density function
2. Cumulative distribution function
3. Expected value
4. Normal distribution
5. Exponential and Gamma distribution

Cumulative distribution functions and expected values

- Cumulative distribution function （CDF)
- Recall the CDF of discrete random variable

Cumulative distribution functions and expected values

- Cumulative distribution function （CDF)
- Recall the CDF of discrete random variable
  - $F_X(x) = P(X \leq x) = \sum_{i=1}^{x} P(X = i)$

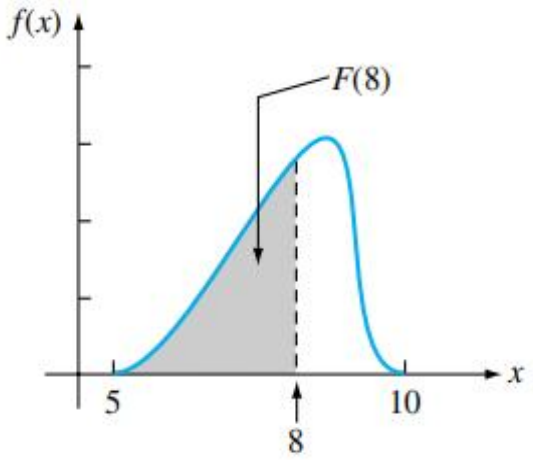
Cumulative distribution functions and expected values

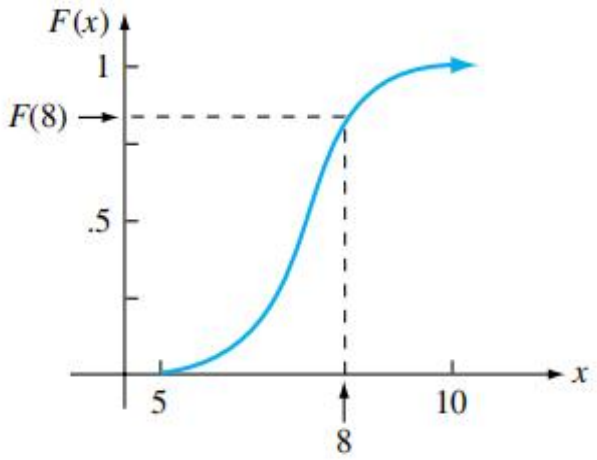
- Cumulative distribution function （CDF)
- Recall the CDF of discrete random variable
  - $F_X(x) = P(X \leq x) = \sum_{i=1}^{x} P(X = i)$
- CDF of continuous random variable
  - $F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(x)dx$

华南师范大學

Cumulative distribution functions and expected values

- Cumulative distribution function （CDF)
- Recall the CDF of discrete random variable
  - $F_X(x) = P(X \leq x) = \sum_{i=1}^{x} P(X = i)$
- CDF of continuous random variable
  - $F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(x)dx$
- For each *x*, $F_X(x)$ is the area under the PDF curve to the left of *x*





Cumulative distribution functions and expected values

- PDF derivation from CDF
- For a discrete random variable recall how we obtained PMF from CDF?

Cumulative distribution functions and expected values

- PDF derivation from CDF
- For a discrete random variable recall how we obtained PMF from CDF?
  - $P(X = a) = F_X(a) - F_X(a - 1)$

Cumulative distribution functions and expected values

- PDF derivation from CDF
- For a discrete random variable recall how we obtained PMF from CDF?
  - $P(X = a) = F_X(\text{a}) - F_X(\text{a} - 1)$
- Similarly, we can obtain PDF from CDF as follows
  - $f_X(x) = \frac{d}{dx} F_X(x)$

華南師範大學

Cumulative distribution functions and expected values

- Consider the function

$$g(x) = \begin{cases} c & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

- For what value of c is $g(x)$ a legitimate PDF?

Cumulative distribution functions and expected values

- Consider the function

$$g(x) = \begin{cases} c & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

- For what value of c is $g(x)$ a legitimate PDF?

$$\int_a^b c\,dx = [cx]_a^b = c(b-a) = 1$$

This implies that $c = \dfrac{1}{b-a}$.

Cumulative distribution functions and expected values

- Consider the function

$$g(x) = \begin{cases} c & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

- For what value of c is $g(x)$ a legitimate PDF?

$$\int_a^b c\,dx = [cx]_a^b = c(b-a) = 1$$

$$\text{This implies that } c = \frac{1}{b-a}.$$

- Find the CDF

Cumulative distribution functions and expected values

- Consider the function

$$g(x) = \begin{cases} c & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

- For what value of c is $g(x)$ a legitimate PDF?

$$\int_a^b c\,dx = [cx]_a^b = c(b-a) = 1$$

This implies that $c = \dfrac{1}{b-a}$.

- Find the CDF

$$F_X(x) = \int_a^x \frac{du}{b-a}$$

$$= \begin{cases} 0 & x < a \\ \dfrac{x-a}{b-a} & a \le x < b \\ 1 & x \ge b \end{cases}$$

華南師範大學

Cumulative distribution functions and expected values

- Consider the function

$$f(x) = \begin{cases} 2x & 0 \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

- For what value of b, is $f(x)$ a legitimate PDF

Cumulative distribution functions and expected values

- Consider the function

$$f(x) = \begin{cases} 2x & 0 \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

- For what value of b, is $f(x)$ a legitimate PDF

$$\int_0^b 2x\,dx = [x^2]\Big|_0^b = b^2 = 1$$

Thus, $b = 1$.

Cumulative distribution functions and expected values

- Consider the function

$$f(x) = \begin{cases} 2x & 0 \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

- For what value of b, is $f(x)$ a legitimate PDF

$$\int_0^b 2x\,dx = [x^2]\Big|_0^b = b^2 = 1$$

Thus, $b = 1$.

- Find the CDF

Cumulative distribution functions and expected values

- Consider the function

$$f(x) = \begin{cases} 2x & 0 \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

- For what value of b, is $f(x)$ a legitimate PDF

$$\int_0^b 2x dx = [x^2]\Big|_0^b = b^2 = 1$$

Thus, $b = 1$.

- Find the CDF

$$F(x) = \int_{-\infty}^{x} f(u)du = \begin{cases} 0 & x < 0 \\ x^2 & 0 \le x < 1 \\ 1 & x \ge 1 \end{cases}$$

華南師範大學

Cumulative distribution functions and expected values

- The PDF of the time T it takes a bank teller to serve a customer is defined by

$$f_T(t) = \begin{cases} \frac{1}{6} & 2 \le t \le 8 \\ 0 & \text{otherwise} \end{cases}$$

- What is the CDF of T?

Cumulative distribution functions and expected values

- The PDF of the time T it takes a bank teller to serve a customer is defined by

$$f_T(t) = \begin{cases} \frac{1}{6} & 2 \le t \le 8 \\ 0 & \text{otherwise} \end{cases}$$

- What is the CDF of T?

$$F_T(t) = P[T \le t] = \int_{-\infty}^{t} f_T(u)du$$

$$= \int_2^t \frac{1}{6}du = \left[\frac{u}{6}\right]_2^t$$

$$= \begin{cases} 0 & t < 2 \\ \frac{t-2}{6} & 2 \le t < 8 \\ 1 & t \ge 8 \end{cases}$$

Cumulative distribution functions and expected values

- The PDF of the time T it takes a bank teller to serve a customer is defined by

$$f_T(t) = \begin{cases} \frac{1}{6} & 2 \le t \le 8 \\ 0 & \text{otherwise} \end{cases}$$

- What is the probability that a customer is served in less than 5 minutes?

Cumulative distribution functions and expected values

- The PDF of the time T it takes a bank teller to serve a customer is defined by

$$f_T(t) = \begin{cases} \frac{1}{6} & 2 \le t \le 8 \\ 0 & \text{otherwise} \end{cases}$$

- What is the probability that a customer is served in less than 5 minutes?

$$P[T < 5] = F_T(5) = \frac{5-2}{6} = 0.5$$

Cumulative distribution functions and expected values

- The CDF of the random variable X is defined by

$$F_X(x) = \begin{cases} 0 & x < 2 \\ A(x-2) & 2 \le x < 6 \\ 1 & x \ge 6 \end{cases}$$

- What is the value of A?

Cumulative distribution functions and expected values

- The CDF of the random variable X is defined by

$$F_X(x) = \begin{cases} 0 & x < 2 \\ A(x-2) & 2 \le x < 6 \\ 1 & x \ge 6 \end{cases}$$

- What is the value of A?
- To find A, we know that $F_X(6) = 1$. Thus, from the definition of the CDF we have

$$F_X(6) = A(6-2) = 4A = 1 \Rightarrow A = \frac{1}{4}$$

華南師範大學

Cumulative distribution functions and expected values

- The CDF of the random variable X is defined by

$$F_X(x) = \begin{cases} 0 & x < 2 \\ A(x-2) & 2 \le x < 6 \\ 1 & x \ge 6 \end{cases}$$

- The probability that X lies between 3 and 5?

Cumulative distribution functions and expected values

- The CDF of the random variable X is defined by

$$F_X(x) = \begin{cases} 0 & x < 2 \\ A(x-2) & 2 \le x < 6 \\ 1 & x \ge 6 \end{cases}$$

- The probability that X lies between 3 and 5?

$$P[3 \le X \le 5] = F_X(5) - F_X(3) = \frac{1}{4}\{(5-2) - (3-2)\} = \frac{1}{2}$$

Cumulative distribution functions and expected values

- The CDF of the random variable X is defined by

$$F_X(x) = \begin{cases} 0 & x < 2 \\ A(x-2) & 2 \le x < 6 \\ 1 & x \ge 6 \end{cases}$$

- The probability that X lies between 3 and 5?

$$P[3 \le X \le 5] = F_X(5) - F_X(3) = \frac{1}{4}\{(5-2) - (3-2)\} = \frac{1}{2}$$

- What is its PDF?

Cumulative distribution functions and expected values

- The CDF of the random variable X is defined by

$$F_X(x) = \begin{cases} 0 & x < 2 \\ A(x-2) & 2 \le x < 6 \\ 1 & x \ge 6 \end{cases}$$

- The probability that X lies between 3 and 5?

$$P[3 \le X \le 5] = F_X(5) - F_X(3) = \frac{1}{4}\{(5-2) - (3-2)\} = \frac{1}{2}$$

- What is its PDF?

$$f_X(x) = \frac{d}{dx}F_X(x) = \begin{cases} A & 2 \le x \le 6 \\ 0 & \text{otherwise} \end{cases}$$

華南師範大學

1. Probability density function
2. Cumulative distribution function
3. Expected value
4. Normal distribution
5. Exponential and Gamma distribution

Expected value

- Expected value
- Recall expected value of discrete random variable

Expected value

- Expected value
- Recall expected value of discrete random variable
  - $E(X) = \sum_x x * p_X(x)$

Expected value

- Expected value
- Recall expected value of discrete random variable
  - $E(X) = \sum_x x * p_X(x)$
- Expected value of continuous random variable
  - $E(x) = \int_{-\infty}^{\infty} x * f_X(x)dx$

華南師範大學

Expected value

- The PDF of continuous random variable X is:

$$f(x) = \begin{cases} \frac{3}{2}(1 - x^2) & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

- What is its expected value?

Expected value

- The PDF of continuous random variable X is:

$$f(x) = \begin{cases} \frac{3}{2}(1 - x^2) & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

- What is its expected value?

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx = \int_0^1 x \cdot \frac{3}{2}(1 - x^2)\, dx$$

$$= \frac{3}{2}\int_0^1 (x - x^3)\, dx = \frac{3}{2}\left(\frac{x^2}{2} - \frac{x^4}{4}\right)\Bigg|_{x=0}^{x=1} = \frac{3}{8}$$

Expected value

- Find the expected value of the random variable X whose PDF is

$$f_X(x) = \begin{cases} 0 & x < a \\ \dfrac{1}{b-a} & a \le x \le b \\ 0 & x > b \end{cases}$$

Expected value

- Find the expected value of the random variable X whose PDF is

$$f_X(x) = \begin{cases} 0 & x < a \\ \dfrac{1}{b-a} & a \le x \le b \\ 0 & x > b \end{cases}$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx = \int_a^b \frac{x}{b-a}\,dx = \left[\frac{x^2}{2(b-a)}\right]_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)}$$

$$= \frac{b+a}{2}$$

Expected value

- Find the expected value of the random variable X whose PDF is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda_x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Expected value

- Find the expected value of the random variable X whose PDF is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda_x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

The expected value of X is given by

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^{\infty} x\lambda e^{-\lambda x} dx$$

Let $dv = \lambda e^{-\lambda x} dx$ and $u = x$. This means that $v = -e^{-\lambda x}$ and $du = dx$. Thus, integrating by parts, we obtain

$$E[X] = [uv]_0^{\infty} - \int_0^{\infty} v du$$

$$= [-xe^{-\lambda x}]_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx = 0 - \left[\frac{e^{-\lambda x}}{\lambda}\right]_0^{\infty}$$

$$= \frac{1}{\lambda}$$

Expected value

- Expected value of function of X
- For discrete random variable X, if $Y = h(X)$, *recall* what is $E(Y)$?

華南師範大學

Expected value

- Expected value of function of X
- For discrete random variable X, if $Y = h(X)$, *recall* what is $E(Y)$?
  - $\mathrm{E(Y)} = \sum_y y * p(y) = \sum_x h(x) * p(x)$

Expected value

- Expected value of function of X
- For discrete random variable X, if $Y = h(X)$, *recall* what is $E(Y)$?
  - $\mathrm{E(Y)} = \sum_y y * p(y) = \sum_x h(x) * p(x)$
- For continuous random variable
  - $\mathrm{E(Y)} = \int_{-\infty}^{\infty} y * f_Y(y)dy = \int_{-\infty}^{\infty} h(x) * f_X(x)dx$

華南師範大學

Expected value

- Expected value of linear function of X
- For discrete random variable X, if $Y = h(X) = a*X + b$, what is $E(Y)$?

华南师范大学

Expected value

- Expected value of linear function of X
- For discrete random variable X, if $Y = h(X) = a*X + b$, what is $E(Y)$?
  - $E(Y) = a * E(X) + b$

Expected value

- Expected value of linear function of X
- For discrete random variable X, if $Y = h(X) = a*X + b$, what is $E(Y)$?
  - $E(Y) = a * E(X) + b$
- For continuous random variable
  - $E(Y) = a * E(X) + b$

$$E[Y] = E[aX + b] = \int_{-\infty}^{\infty} (ax + b) f_X(x) dx = \int_{-\infty}^{\infty} axf_X(x) dx + \int_{-\infty}^{\infty} bf_X(x) dx$$

$$= a \int_{-\infty}^{\infty} xf_X(x) dx + b \int_{-\infty}^{\infty} f_X(x) dx$$

$$= aE[X] + b$$

1. Probability density function
2. Cumulative distribution function
3. Expected value
4. Normal distribution
5. Exponential and Gamma distribution

Normal distribution

- Normal random variable (正态随机变量)
- Normal random variable or Gaussian random variable X with parameter $\mu, -\infty < \mu < \infty$, and $\sigma, 0 < \sigma$, is denoted as $X \sim N(\mu, \sigma^2)$ and has the following PDF

$$f(x;\, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-(x-\mu)^2/(2\sigma^2)} \qquad -\infty < x < \infty$$

- The CDF is

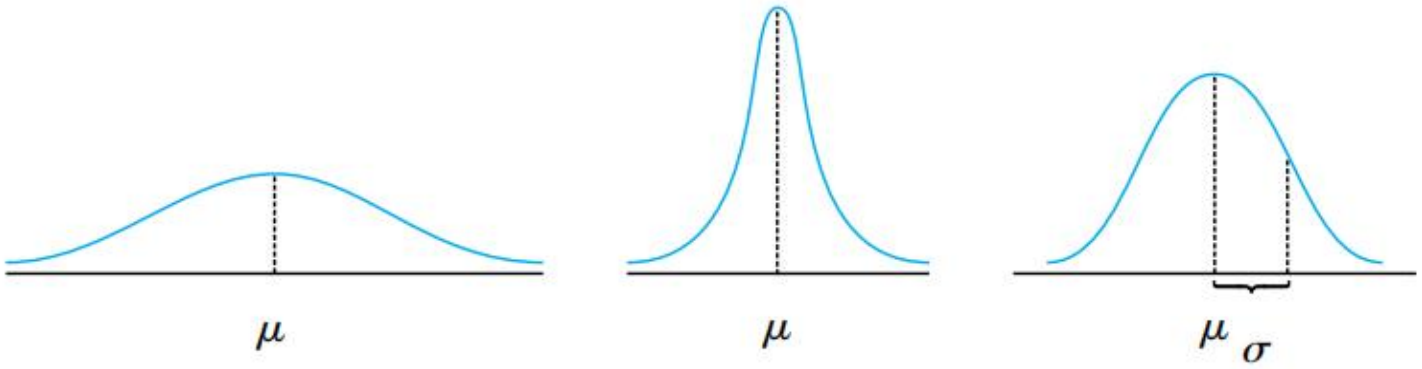
$$F_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \int_{-\infty}^{x} e^{-(q-\mu)^2/(2\sigma^2)} dq$$

- E[X] = μ, V[X] = σ^2

華南師範大學

Normal distribution

- PDF curve of normal random variable is symmetric about expected value μ and bell-shaped, so the center of the curve is μ
- Large value of σ makes the graph spread, but small value of σ makes a high peak above μ



Normal distribution

- Standard normal random variable (标准正态随机变量)
- A normal random variable with $\mu = 1, \sigma = 0$ is called standard normal random variable, and is denoted by Z
- The PDF of Z is

$$f(z; 0, 1) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \qquad -\infty < z < \infty$$

- The CDF of Z is denoted by $\Phi(z)$
- Appendix Table A.3 of text book gives the values of $\Phi(z)$

華南師範大學

Normal distribution

- Standard normal random variable (标准正态随机变量)
- Let $Y = (X - \mu)/\sigma$, then Y is a standard normal random variable
- Hence, we can obtain CDF of X by CDF of Y, $F_X(x) = F_Y((x - \mu)/\sigma)$
$= \Phi((x - \mu)/\sigma)$

华南师范大學

Normal distribution

- Standard normal random variable (标准正态随机变量)
- If X is a normal random variable with mean $\mu_X = 3$ and variance $\sigma_X{}^2 = 9$, find the probability that X lies between 2 and 5

$$
\begin{aligned}
P[2 < X < 5] &= F_X(5) - F_X(2) \\
&= \Phi\left(\frac{5-\mu_X}{\sigma_X}\right) - \Phi\left(\frac{2-\mu_X}{\sigma_X}\right) \\
&= \Phi\left(\frac{5-3}{3}\right) - \Phi\left(\frac{2-3}{3}\right) = \Phi\left(\frac{2}{3}\right) - \Phi\left(-\frac{1}{3}\right) \\
&= \Phi\left(\frac{2}{3}\right) - \left\{1 - \Phi\left(\frac{1}{3}\right)\right\} \\
&= \Phi\left(\frac{2}{3}\right) + \Phi\left(\frac{1}{3}\right) - 1 \\
&= 0.7486 + 0.6293 - 1 = 0.3779
\end{aligned}
$$

華南師範大學

Exercise

- Prove that for a normal random variable X with mean μ and variance σ^2, we have $E[X] = \mu$, $V[X] = \sigma^2$
- Prove that for the above random variable X, why random variable $Y = (X - \mu)/\sigma$ is a standard normal random variable

Exercise

- Prove that for a normal random variable X with mean μ and variance σ^2, we have $\mathrm{E}[X] = \mu$, $\mathrm{V}[X] = \sigma^2$

$$E[X] = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx.$$

$$z = \frac{x-\mu}{\sigma} \Rightarrow x = \sigma z + \mu.$$

$$E[X] = \int_{-\infty}^{\infty} (\sigma z + \mu) \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz.$$

$$E[X] = \int_{-\infty}^{\infty} \sigma z \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz + \int_{-\infty}^{\infty} \mu \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz.$$

$$\int_{-\infty}^{\infty} \sigma z \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = \sigma \cdot \int_{-\infty}^{\infty} z \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz.$$

Since the integrand $ze^{-z^2/2}$ is an odd function (symmetric around zero), and we're integrating over the symmetric interval $(-\infty, \infty)$, this integral is zero:

$$\int_{-\infty}^{\infty} z \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = 0.$$

$$\int_{-\infty}^{\infty} \mu \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = \mu \cdot \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz.$$

The integral $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = 1$ because it is the integral of a standard normal density over its entire range. Therefore:

$$\int_{-\infty}^{\infty} \mu \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = \mu \cdot 1 = \mu.$$

華南師範大學

Exercise

- Prove that for a normal random variable X with mean μ and variance σ^2, we have $E[X] = \mu$, $V[X] = \sigma^2$

Since $X \sim N(\mu, \sigma^2)$, we can express X as:

$$X = \mu + \sigma Z,$$

where $Z \sim N(0, 1)$, meaning that Z is a standard normal random variable with mean 0 and variance 1.

Now, we want to find $E[X^2]$:

$$E[X^2] = E[(\mu + \sigma Z)^2].$$

Expanding the square:

$$(\mu + \sigma Z)^2 = \mu^2 + 2\mu\sigma Z + \sigma^2 Z^2.$$

Taking the expectation of both sides, we get:

$$E[X^2] = E[\mu^2 + 2\mu\sigma Z + \sigma^2 Z^2].$$

Using the linearity of expectation, this separates as:

$$E[X^2] = E[\mu^2] + E[2\mu\sigma Z] + E[\sigma^2 Z^2].$$

Since μ^2 and σ^2 are constants, we have:

$$E[X^2] = \mu^2 + 2\mu\sigma E[Z] + \sigma^2 E[Z^2].$$

1. **First Term**: $E[\mu^2] = \mu^2$ since μ^2 is a constant.
2. **Second Term**: $2\mu\sigma E[Z]$. Since $Z \sim N(0, 1)$, we know that $E[Z] = 0$. Therefore,

$$2\mu\sigma E[Z] = 2\mu\sigma \cdot 0 = 0.$$

3. **Third Term**: $\sigma^2 E[Z^2]$. To find $E[Z^2]$, recall that for a standard normal random variable $Z \sim N(0, 1)$, the variance $V[Z] = E[Z^2] - (E[Z])^2 = 1$. Since $E[Z] = 0$, we have $E[Z^2] = 1$. Therefore,

$$\sigma^2 E[Z^2] = \sigma^2 \cdot 1 = \sigma^2.$$

Exercise

- Prove that for the above random variable X, why random variable $Y = (X - \mu)/\sigma$ is a standard normal random variable

Given:

- $X \sim N(\mu, \sigma^2)$, meaning X has a mean μ and variance σ^2.
- We define a new random variable $Y = \frac{X-\mu}{\sigma}$.

We need to show that $Y \sim N(0, 1)$, which means:

1. $E[Y] = 0$.
2. $V[Y] = 1$.

Since $Y = \frac{X-\mu}{\sigma}$, we can use the linearity of expectation to find $E[Y]$:

$$E[Y] = E\left[\frac{X-\mu}{\sigma}\right].$$

Since $\frac{1}{\sigma}$ is a constant, it can be factored out:

$$E[Y] = \frac{1}{\sigma}E[X-\mu].$$

Now, using the linearity of expectation again:

$$E[Y] = \frac{1}{\sigma}\left(E[X] - \mu\right).$$

Since $E[X] = \mu$, we have:

$$E[Y] = \frac{1}{\sigma}(\mu - \mu) = \frac{1}{\sigma} \cdot 0 = 0.$$

Thus,

$$E[Y] = 0.$$

To find $V[Y]$, we use the definition of variance:

$$V[Y] = E\left[(Y - E[Y])^2\right].$$

Since $E[Y] = 0$, this simplifies to:

$$V[Y] = E[Y^2].$$

Now, substitute $Y = \frac{X-\mu}{\sigma}$:

$$V[Y] = E\left[\left(\frac{X-\mu}{\sigma}\right)^2\right].$$

This further simplifies to:

$$V[Y] = \frac{1}{\sigma^2}E\left[(X-\mu)^2\right].$$

By definition, $V[X] = E[(X-\mu)^2] = \sigma^2$, so we get:

$$V[Y] = \frac{1}{\sigma^2} \cdot \sigma^2 = 1.$$

Thus,

$$V[Y] = 1.$$

1. Probability density function
2. Cumulative distribution function
3. Expected value
4. Normal distribution
5. Exponential and Gamma distribution

Exponential distribution

- Exponential random variable （指数随机变量）
- The exponential random variable is usually used to model the probability distribution of time between the occurrence of successive events, e.g., customers arriving at a service facility or cars arriving at a gas station
- Exponential random variable has the following PDF

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- What is its CDF?

Exponential distribution

- Exponential random variable （指数随机变量）
- The exponential random variable is usually used to model the probability distribution of time between the occurrence of successive events, e.g., customers arriving at a service facility or cars arriving at a gas station
- Exponential random variable has the following PDF

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- What is its CDF?

$$F_X(x) = P[X \leq x] = \int_0^x f_X(y)dy = \int_0^x \lambda e^{-\lambda y} dy$$
$$= \left[-e^{-\lambda y}\right]_0^x = 1 - e^{-\lambda x}$$

Exponential distribution

- Exponential random variable （指数随机变量）
- $E[X] = 1/\lambda, V[X] = 1/\lambda^2$

华南师范大學

Exponential distribution

- Exponential random variable （指数随机变量）
- $E[X] = 1/\lambda, V[X] = 1/\lambda^2$

$$E[X] = \int_0^\infty x f_X(x) dx = \int_0^\infty x\lambda e^{-\lambda x} dx$$

Let $u = x$ and $dv = \lambda e^{-\lambda x} dx$, which means that $du = dx$ and $v = -e^{-\lambda x}$. Thus, integrating by parts we obtain

$$E[X] = \int_0^\infty x\lambda e^{-\lambda x} dx$$

$$= \left[-xe^{-\lambda x}\right]_0^\infty + \int_0^\infty e^{-\lambda x} dx = 0 - \left[\frac{1}{\lambda} e^{-\lambda x}\right]_0^\infty$$

$$= \frac{1}{\lambda}$$

華南師範大學

Exponential distribution

- Exponential random variable （指数随机变量）
- $E[X] = 1/\lambda, V[X] = 1/\lambda^2$

By repeated use of integration by parts, it can be shown that the nth moment of X is given by

$$E[X^n] = \int_0^\infty x^n \lambda e^{-\lambda x} dx = \frac{n!}{\lambda^n}$$

Thus, the variance of X is given by

$$\sigma_X^2 = E[X^2] - (E[X])^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

华南师范大學

Exponential distribution

- Assume that the length of phone calls made at a particular telephone booth is exponentially distributed with a mean of 3 minutes. If you arrive at the telephone booth just as a person was about to make a call, find the following:
- The probability that you will wait more than 5 minutes before the person is done

Exponential distribution

- Assume that the length of phone calls made at a particular telephone booth is exponentially distributed with a mean of 3 minutes. If you arrive at the telephone booth just as a person was about to make a call, find the following:
- The probability that you will wait more than 5 minutes before the person is done
- Let X be a random variable that denotes the length of calls. Since the mean length of calls is 3, we have $E[X] = 1/\lambda = 3$, then the PDF of X is

$$f_X(x) = \lambda e^{-\lambda x} = \frac{1}{3} e^{-x/3}$$

$$P[X > 5] = \int_5^{\infty} \frac{1}{3} e^{-x/3} dx = \left[-e^{-x/3}\right]_5^{\infty} = e^{-5/3} = 0.1889$$

Exponential distribution

- Assume that the length of phone calls made at a particular telephone booth is exponentially distributed with a mean of 3 minutes. If you arrive at the telephone booth just as a person was about to make a call, find the following:
- The probability that your waiting time is from 2 minutes and 6 minutes

Exponential distribution

- Assume that the length of phone calls made at a particular telephone booth is exponentially distributed with a mean of 3 minutes. If you arrive at the telephone booth just as a person was about to make a call, find the following:
- The probability that your waiting time is from 2 minutes and 6 minutes

$$P[2 \le X \le 6] = \int_2^6 \frac{1}{3} e^{-x/3} dx = \left[-e^{-x/3}\right]_2^6 = F_X(6) - F_X(2)$$
$$= e^{-2/3} - e^{-2} = 0.3781$$

Exponential distribution

- The PDF of a random variable T is given by

$$f_T(t) = ke^{-4t} \quad t \geq 0$$

- What is the value of k?

Exponential distribution

- The PDF of a random variable T is given by

$$f_T(t) = ke^{-4t} \quad t \geq 0$$

- What is the value of k?

$$\int_0^\infty f_T(t)dt = k\int_0^\infty e^{-4t}dt = 1 \Rightarrow k\left[-\frac{e^{-4t}}{4}\right]_0^\infty = \frac{k}{4} = 1 \Rightarrow k = 4$$

Exponential distribution

- The PDF of a random variable T is given by

$$f_T(t) = ke^{-4t} \quad t \geq 0$$

- What is the value of k?

$$\int_0^\infty f_T(t)dt = k\int_0^\infty e^{-4t}dt = 1 \Rightarrow k\left[-\frac{e^{-4t}}{4}\right]_0^\infty = \frac{k}{4} = 1 \Rightarrow k = 4$$

- What is the expected value of T?

Exponential distribution

- The PDF of a random variable T is given by

$$f_T(t) = ke^{-4t} \quad t \geq 0$$

- What is the value of k?

$$\int_0^\infty f_T(t)dt = k\int_0^\infty e^{-4t}dt = 1 \Rightarrow k\left[-\frac{e^{-4t}}{4}\right]_0^\infty = \frac{k}{4} = 1 \Rightarrow k = 4$$

- What is the expected value of T?

$$E[T] = \frac{1}{4}$$

Exponential distribution

- The PDF of a random variable T is given by

$$f_T(t) = ke^{-4t} \quad t \geq 0$$

- What is the value of k?

$$\int_0^\infty f_T(t)dt = k\int_0^\infty e^{-4t}dt = 1 \Rightarrow k\left[-\frac{e^{-4t}}{4}\right]_0^\infty = \frac{k}{4} = 1 \Rightarrow k = 4$$

- What is the expected value of T?

$$E[T] = \frac{1}{4}$$

- What is the probability that T is less than 1

Exponential distribution

- The PDF of a random variable T is given by

$$f_T(t) = ke^{-4t} \quad t \geq 0$$

- What is the value of k?

$$\int_0^\infty f_T(t)dt = k\int_0^\infty e^{-4t}dt = 1 \Rightarrow k\left[-\frac{e^{-4t}}{4}\right]_0^\infty = \frac{k}{4} = 1 \Rightarrow k = 4$$

- What is the expected value of T?

$$E[T] = \frac{1}{4}$$

- What is the probability that T is less than 1

$$P[T<1] = \int_0^1 f_T(t)dt = \int_0^1 4e^{-4t}dt = [-e^{-4t}]_0^1 = 1 - e^{-4} = 0.9817$$

Exponential distribution

- The lifetime X of a system in weeks is given by the following PDF

$$f_X(x) = \begin{cases} 0.25e^{-0.25x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- What is the expected value of X?

Exponential distribution

- The lifetime X of a system in weeks is given by the following PDF

$$f_X(x) = \begin{cases} 0.25e^{-0.25x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- What is the expected value of X?

$$E[X] = 1/0.25 = 4$$

Exponential distribution

- The lifetime X of a system in weeks is given by the following PDF

$$f_X(x) = \begin{cases} 0.25e^{-0.25x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- What is the expected value of X?

$$E[X] = 1/0.25 = 4$$

- What is the CDF of X?

Exponential distribution

- The lifetime X of a system in weeks is given by the following PDF

$$f_X(x) = \begin{cases} 0.25e^{-0.25x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- What is the expected value of X?

$$E[X] = 1/0.25 = 4$$

- What is the CDF of X?

$$F_X(x) = 1 - e^{-0.25x} \qquad x \geq 0$$

Exponential distribution

- The lifetime X of a system in weeks is given by the following PDF

$$f_X(x) = \begin{cases} 0.25e^{-0.25x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- What is the expected value of X?

$$E[X] = 1/0.25 = 4$$

- What is the CDF of X?

$$F_X(x) = 1 - e^{-0.25x} \qquad x \geq 0$$

- What is the probability that the system will not fail within two weeks?

華南師範大學

Exponential distribution

- The lifetime X of a system in weeks is given by the following PDF

$$f_X(x) = \begin{cases} 0.25e^{-0.25x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- What is the expected value of X?

$$E[X] = 1/0.25 = 4$$

- What is the CDF of X?

$$F_X(x) = 1 - e^{-0.25x} \qquad x \geq 0$$

- What is the probability that the system will not fail within two weeks?

$$P[X > 2] = 1 - F_X(2) = e^{-0.25(2)} = e^{-0.5} = 0.6065$$

Exponential distribution

- Forgetfulness property of exponential distribution （无记忆性性质）
- Recall forgetfulness property of geometric distribution

华南师范大學

Exponential distribution

- Forgetfulness property of exponential distribution （无记忆性性质）
- Recall forgetfulness property of geometric distribution
- Let X be a random variable that denotes the number of trials until the first success. Suppose we have observed a number n of these trials, and they are all failures. We would like to know the probability of number K of additional trials until the first success?

$$p_K(k) = p(1-p)^{k-1}$$

华南师范大學

Exponential distribution

- Forgetfulness property of exponential distribution （无记忆性性质）
- Recall forgetfulness property of geometric distribution
- Let X be a random variable that denotes the number of trials until the first success. Suppose we have observed a number n of these trials, and they are all failures. We would like to know the probability of number K of additional trials until the first success?

$$p_K(k) = p(1-p)^{k-1}$$

- Suppose the life X of an equipment is exponentially distributed with a mean of $1/\lambda$. Assume that the equipment has not failed by time t. What is the probability of additional life time denoted by s?

Exponential distribution

- Forgetfulness property of exponential distribution （无记忆性性质）
- Recall forgetfulness property of geometric distribution
- Let X be a random variable that denotes the number of trials until the first success. Suppose we have observed a number n of these trials, and they are all failures. We would like to know the probability of number K of additional trials until the first success?

$$p_K(k) = p(1-p)^{k-1}$$

- Suppose the life X of an equipment is exponentially distributed with a mean of $1/\lambda$. Assume that the equipment has not failed by time t. What is the probability of additional life time denoted by s?

$$P[X \le s+t | X > t] = \frac{P[(X \le s+t) \cap (X > t)]}{P[X > t]}$$

$$= \frac{P[t < X \le s+t]}{P[X > t]} = \frac{F_X(s+t) - F_X(t)}{1 - F_X(t)}$$

$$= \frac{\left[1 - e^{-\lambda(s+t)}\right] - \left[1 - e^{\lambda t}\right]}{e^{-\lambda t}} = \frac{e^{-\lambda t} - e^{-\lambda(s+t)}}{e^{-\lambda t}}$$

$$= 1 - e^{-\lambda s} = F_X(s) = P[X \le s]$$

Exponential distribution

- Forgetfulness property of exponential distribution （无记忆性性质）
- Recall forgetfulness property of geometric distribution
- Let X be a random variable that denotes the number of trials until the first success. Suppose we have observed a number n of these trials, and they are all failures. We would like to know the probability of number K of additional trials until the first success?

$$p_K(k) = p(1-p)^{k-1}$$

- Suppose the life X of an equipment is exponentially distributed with a mean of $1/\lambda$. Assume that the equipment has not failed by time t. What is the probability of additional life time denoted by s?

$$P[X \le s+t | X > t] = \frac{P[(X \le s+t) \cap (X > t)]}{P[X > t]}$$

$$= \frac{P[t < X \le s+t]}{P[X > t]} = \frac{F_X(s+t) - F_X(t)}{1 - F_X(t)}$$

$$= \frac{\left[1 - e^{-\lambda(s+t)}\right] - \left[1 - e^{\lambda t}\right]}{e^{-\lambda t}} = \frac{e^{-\lambda t} - e^{-\lambda(s+t)}}{e^{-\lambda t}}$$

$$= 1 - e^{-\lambda s} = F_X(s) = P[X \le s]$$

Original exponential distribution

Exponential distribution

- The lifetime X of a system in weeks is given by the following PDF

$$f_X(x) = \begin{cases} 0.25e^{-0.25x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- Given that the system has not failed by the end of the fourth week, what is the probability that it will fail between the fourth and sixth weeks?

Exponential distribution

- The lifetime X of a system in weeks is given by the following PDF

$$f_X(x) = \begin{cases} 0.25e^{-0.25x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- Given that the system has not failed by the end of the fourth week, what is the probability that it will fail between the fourth and sixth weeks?
- Because of the forgetfulness property of the exponential distribution, the probability that the system will fail between the fourth and sixth weeks, given that is has not failed by the end of the fourth week, is simply the probability that it will fail within 2 weeks. That is,

$$P[4 < X < 6 | X > 4] = P[X \leq 2] = F_X(2) = 1 - e^{-0.5} = 0.3935$$

Exponential distribution

- Relationship between exponential and poisson distributions
- Let λ denote the average number of Poisson arrivals per second. Then the average number of arrivals in t seconds is λt. If K denotes this Poisson random variable, the PMF of the number of arrivals during t seconds is

Exponential distribution

- Relationship between exponential and poisson distributions
- Let λ denote the average number of Poisson arrivals per second. Then the average number of arrivals in t seconds is λt. If K denotes this Poisson random variable, the PMF of the number of arrivals during t seconds is

$$p_K(k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t} \quad k = 0, 1, 2, \ldots; \; t \geq 0$$

Exponential distribution

- Relationship between exponential and poisson distributions
- Let λ denote the average number of Poisson arrivals per second. Then the average number of arrivals in t seconds is λt. If K denotes this Poisson random variable, the PMF of the number of arrivals during t seconds is

$$p_K(k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t} \quad k = 0, 1, 2, \ldots; \; t \geq 0$$

- The probability that there is no arrival in an interval of t seconds is $p_K(0) = e^{-\lambda t}$. Thus, the probability that there is at least one arrival in an interval of t seconds is $1 - e^{-\lambda t}$.

華南師範大學

Exponential distribution

- Relationship between exponential and poisson distributions
- Let λ denote the average number of Poisson arrivals per second. Then the average number of arrivals in t seconds is λt. If K denotes this Poisson random variable, the PMF of the number of arrivals during t seconds is

$$p_K(k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t} \quad k = 0, 1, 2, \ldots; \; t \geq 0$$

- The probability that there is no arrival in an interval of t seconds is $p_K(0) = e^{-\lambda t}$. Thus, the probability that there is at least one arrival in an interval of t seconds is $1 - e^{-\lambda t}$.
- But for an exponential random variable Y with parameter λ, the probability that an event occurs no later than time t is given by

$$P[Y \leq t] = F_Y(t) = 1 - e^{-\lambda t}$$

- Thus, an exponential distribution Y with parameter λ describes the intervals between occurrence of events defined by a Poisson random variable K with mean λt

華南師範大學

Exponential distribution

- Relationship between exponential and poisson distributions
- If we define λ as the average number of Poisson arrivals per unit time, it becomes clear why the mean of Y is $1/\lambda$ time units
- If intervals between events in a given process can be modeled by an exponential random variable, then the number of those events that occur during a specific time interval can be modeled by a Poisson random variable

華南師範大學

Exponential distribution

- Relationship between exponential and poisson distributions
- If we define λ as the average number of Poisson arrivals per unit time, it becomes clear why the mean of Y is $1/\lambda$ time units
- If intervals between events in a given process can be modeled by an exponential random variable, then the number of those events that occur during a specific time interval can be modeled by a Poisson random variable
- Similarly, if the number of events that occur during a specific time interval can be modeled by a Poisson random variable, then the interval between successive events can be modeled by an exponential random variable

華南師範大學

Gamma distribution

- Gamma function（伽玛函数）

For $\alpha > 0$, the **gamma function** $\Gamma(\alpha)$ is defined by

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx$$

1. For any $\alpha > 1$, $\Gamma(\alpha) = (\alpha - 1) \cdot \Gamma(\alpha - 1)$
2. For any positive integer, n, $\Gamma(n) = (n - 1)!$
3. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

華南師範大學

Gamma distribution

- Gamma function（伽玛函数）

For any $\alpha > 1$, $\Gamma(\alpha) = (\alpha - 1) \cdot \Gamma(\alpha - 1)$

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t}\, dt$$

To use integration by parts, let:

- $u = t^{\alpha-1}$ (then $du = (\alpha - 1)t^{\alpha-2}\, dt$)
- $dv = e^{-t}\, dt$ (then $v = -e^{-t}$)

Applying integration by parts:

$$\Gamma(\alpha) = \left[-t^{\alpha-1}e^{-t}\right]_0^\infty + (\alpha - 1)\int_0^\infty t^{\alpha-2}e^{-t}\, dt$$

The term $\left[-t^{\alpha-1}e^{-t}\right]_0^\infty$ vanishes because $t^{\alpha-1}e^{-t} \to 0$ as $t \to \infty$ and at $t = 0$. So we get:

$$\Gamma(\alpha) = (\alpha - 1)\int_0^\infty t^{\alpha-2}e^{-t}\, dt$$

Recognizing that the remaining integral is $\Gamma(\alpha - 1)$, we have:

$$\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$$

華南師範大學

Gamma distribution

- Gamma function（伽玛函数）

For any positive integer n, $\Gamma(n) = (n-1)!$

This property can be proven by using the recurrence relation from Property 1.

Starting with $n = 1$:

$$\Gamma(1) = \int_0^\infty e^{-t}\,dt = 1$$

which matches $0! = 1$.

Now, if we assume $\Gamma(k) = (k-1)!$ holds for some integer k, we can use the recurrence relation to prove it for $k+1$:

$$\Gamma(k+1) = k \cdot \Gamma(k) = k \cdot (k-1)! = k!$$

Thus, by induction, $\Gamma(n) = (n-1)!$ for any positive integer n.

華南師範大學

Gamma distribution

- Gamma function（伽玛函数）

$$\Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi}$$

This property requires a different approach and can be proven using the Gaussian integral.

Consider the integral of e^{-x^2} over the entire real line:

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

Then, using the substitution $t = x^2$ (so $dt = 2x\,dx$ or $dx = \frac{dt}{2\sqrt{t}}$) in the integral from 0 to ∞, we get:

$$\int_{0}^{\infty} e^{-t} t^{-\frac{1}{2}}\,dt = \sqrt{\pi}$$

This is precisely the definition of $\Gamma\left(\tfrac{1}{2}\right)$, thus:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

华南师范大學

Gamma distribution

- Gamma random variable （伽玛随机变量）

A continuous random variable X is said to have a gamma distribution if the pdf of X is

$$f_X(x) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where the parameters α and β satisfy $\alpha > 0, \beta > 0$. The **standard gamma distribution** has $\beta = 1$, so the pdf of a standard gamma random variable is given by

$$f_X(x) = \begin{cases} \frac{1}{\Gamma(\alpha)} x^{\alpha-1} e^{-x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Gamma distribution

- Gamma random variable （伽玛随机变量）
- $E[X] = \alpha\beta, V[X] = \alpha\beta^2$